

Advanced Medallion Pipeline & Feature Store Audit

Defensive Data Engineering, Forensic EDA, and ML-Ready
Transformation Lifecycle

| Objectives & Pipeline Architecture



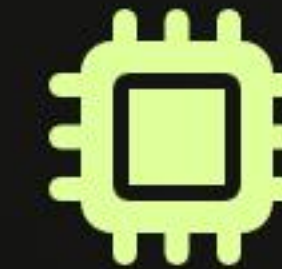
Defensive Auditing

Establishing a "Zero-Trust" ingestion framework to identify and neutralize synthetic data corruption and malformed relational keys.



Medallion Lifecycle

Orchestrating the transition from raw Bronze ingestion to scrubbed Silver validation, culminating in high-performance Gold feature stores.



ML Validation

Closing the loop by feeding curated features into a Random Forest Classifier to validate data readiness and detect target proxy leakage.

SECTION 01

The Forensic Data Audit

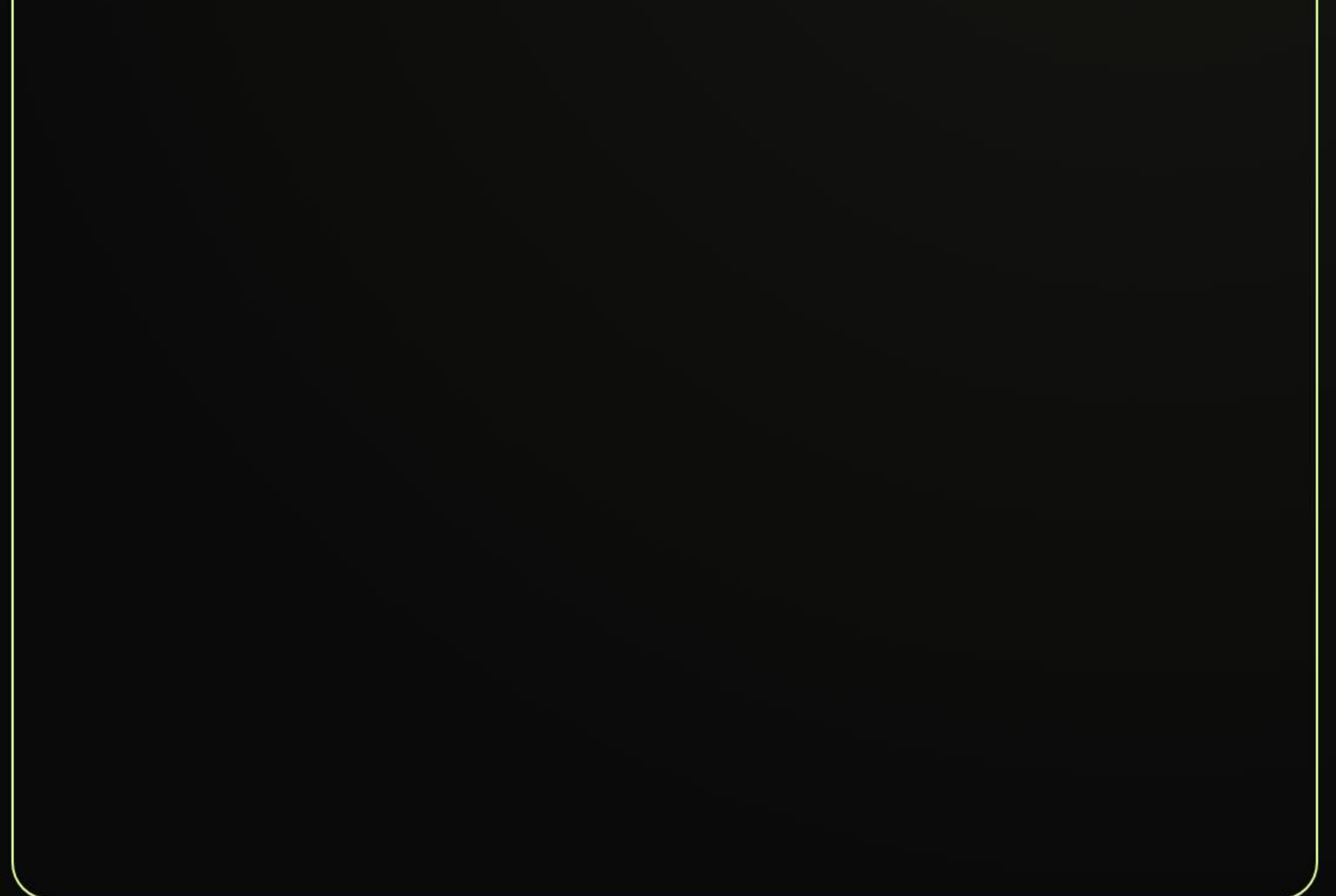
Exposing the anatomy of sabotaged data and synthetic anomalies.

| Master Key Forensics: The Multiplier Effect

Relational integrity is the bedrock of the pipeline. Our audit exposed a critical discrepancy in **Customer_ID** integrity:

- ✓ **Primary Tables:** Exactly 756 malformed 9-character IDs in Attributes and Financials.
- ✓ **The Clickstream Multiplier:** A staggering 13,152 malformed rows identified in high-velocity logs.
- ✓ **Risk Mitigation:** Joining without strict ID filtering would have injected thousands of "ghost" customer profiles.

 Data fragmentation visualization



| Universal Poison Sweep: Category Audit



Garbage Symbols

Detected 9,683 rows in **Type_of_Loan** and 998 rows in **Payment_Behaviour** containing nonsensical ASCII character strings.



Fake Nulls

Identified "NM", "NA", and "_" strings being used as placeholders in categorical columns, necessitating mapping to "Unknown".



Concatenation

Exposed 180-character strings in loan fields where multiple categories were mashed together, creating a cardinality explosion.

| Numeric Reality Check: Extreme Outliers

8,678

Maximum Customer Age

2.38e7

Max Annual Income (\$)

-1

Min Bank Accounts

"Outliers of this magnitude indicate intentional synthetic corruption or severe ingestion failure, requiring capped reality-checks in the Silver layer."

| Transformation Lifecycle: Medallion Flow

01 | Bronze

Raw Ingestion. 215k rows.
Preserves all 'poison' for
historical audit and re-
processing capability.

02 | Silver

Validated Layer. ID filtering,
Regex scrubbing, and
outlier capping. Schema-
resilient logic.

03 | Gold

Analytics Ready.
Denormalized Feature
Store. Parquet format
optimized for distributed ML
training.

Silver Layer: The Cleaner

- ✓ **Relational Filtering:** Strict 10-char enforcement on IDs.
 - ✓ **Regex Scrubbing:** targeted removal of underscores from numeric strings before type-casting.
 - ✓ **Reality Capping:** Truncating impossible Age and Bank Account values to NULL/Cap.
 - ✓ **Deduplication:** Preventing Cartesian explosions in high-velocity click logs.
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Gold Layer: Feature Store Matrix

Feature Class	Column Name	Data Type	Refinement Strategy
Demographics	Age, Occupation	Int / String	Capped outliers, standardized labels
Financials	Annual_Income, Debt	Double	Regex scrubbed, null-handled
Behaviors	Credit_Mix, Payment	String	"Unknown" mapping for fake nulls
Aggregates	Monthly_Clicks	Int	Deduped and joined from Clickstream

| ML Validation: Classifier Performance

Baseline (Leaky) Model

97.28%

Strict (No Leakage) Model

96.85%

Key Validation Metric

The model successfully trained on the **Gold Feature Store** without runtime exceptions, validating the Silver layer's cleaning efficacy.



| Critical Insight: Synthetic Determinism

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"97% accuracy on baseline features is a red flag. The dataset was generated via deterministic if/else logic rather than stochastic human behavior."

— *Forensic Analysis Recommendation*

Our audit of **Feature Importance** revealed that the professor's script used rigid boundaries for bank accounts and income to determine credit mix, lacking real-world variance.

Questions?

Advanced Feature Store & Medallion Pipeline v1.0

Access Project Repository:

<https://github.com/USER/Feature-Store-Pipeline>

| Image Sources



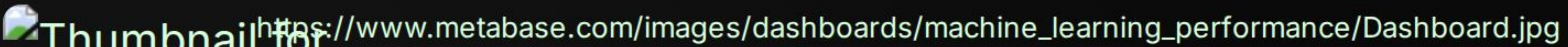
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